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PHNOM PENH INTERNATIONAL UNIVERSITY

HOTEL BOOKING CANCELLATION PREDICTION

Subject: Machine Learning

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Overview

1 The project applies supervised machine learning to predict hotel booking cancellation

2 Dataset booking.csv — 36,285 bookings, 17 feature, 2 classes (canceled / not canceled)

3 Four model implemented : logistic regression , random forest classifier , **Decision Tree and K-Nearest Neighbors(KNN)**

4 Four model tuned using GridSearchCV with stratified cross-validation

5 Evaluated using accuracy, procision, recall, f1 score, confusion matrix and Roc curve

6 Key drivers identified: lead time, special request , market segment and average price

TABLE CONTENT

1. Introduction

Hotel Booking Cancellation Prediction

"Hotel cancellations cost businesses money and mess up planning. My project predicts if a booking will be canceled, before it happens, using real data from over 36,000 hotel bookings. I looked at things like booking lead time, price, and customer type. Bookings made far in advance cancel much more often, while guests with special requests almost never cancel. The goal is to help hotels act early, instead of getting surprised."

2. Data Description

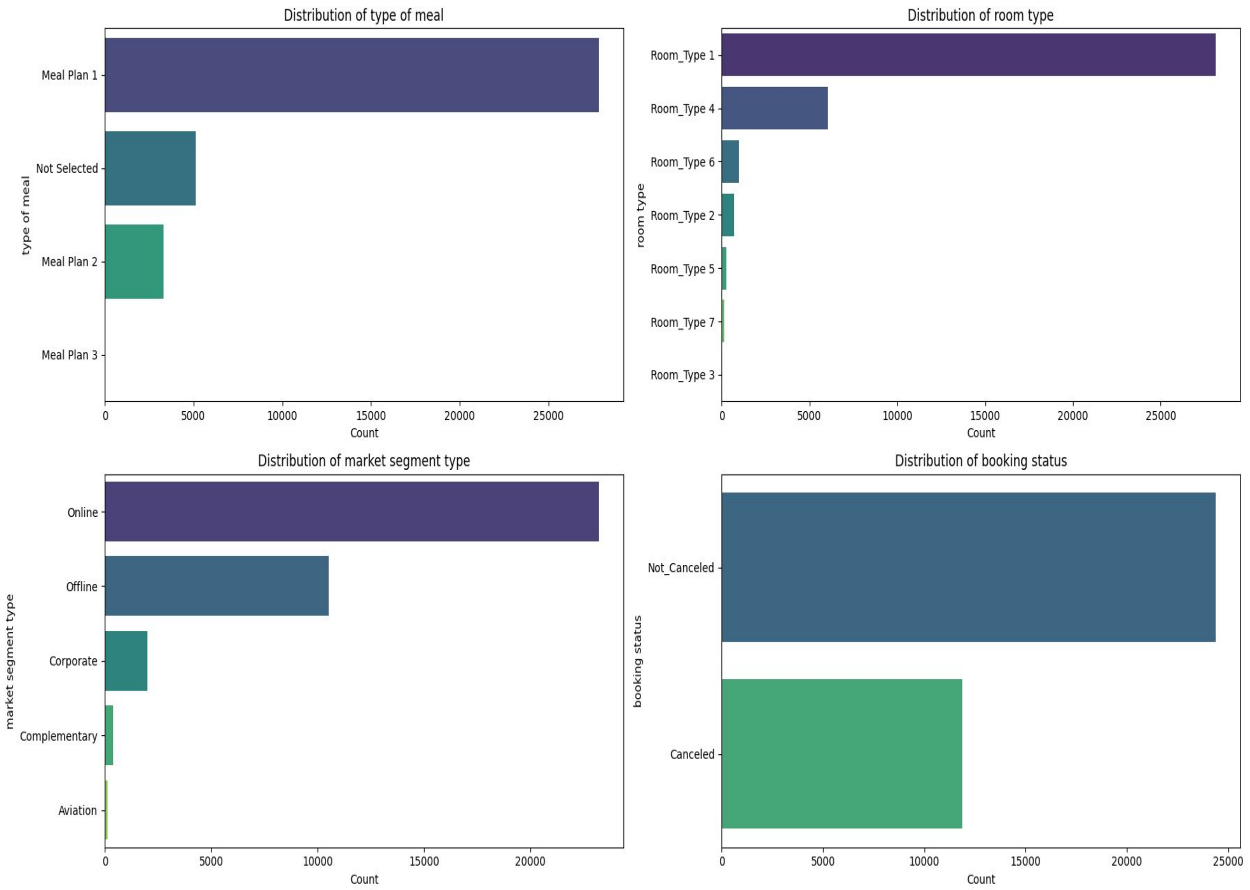
Scope & Dataset Specs

- **Dataset: 36285 rows and 17 columns.**
- **There don’t have any missing value and duplication rows**
- **Target Variable: Booking_Status**

3. Exploratory Data Analysis (EDA)

Analyze the dataset

- Understand booking patterns.
- Explore the distribution of categorical variables.
- Identify factors related to booking cancellations.
- Detect trends and relationships between variables.
- Discover patterns before building machine learning models.



4. Data Processing

1. Encoding Categorical variables

- Separate target value (booking_status).
- Drop irrelevant identifier columns like booking_ID and date of reservation which requires specific handling or feature engineering not directly part of standard encoding for nominal categories.

2. Feature Scaling

Re-apply StandardScaler to ensure numerical features have a mean of 0 and a standard deviation of 1.

3. Feature Select

To help in reducing overfitting, improving accuracy, and reducing training time.

4. Split Feature and Target & train/test split

Split the data into training and testing sets to evaluate our model's performance on unseen data. A common split ratio is 80% for training and 20% for testing.

5. Methodology: Model Algorithms Train Evaluation

Logistic Regression

```
--- Logistic Regression Model Evaluation ---
Accuracy: 0.8057
Precision: 0.7526
Recall: 0.6064
F1-Score: 0.6716
ROC AUC Score: 0.8670
```

Decision Tree

```
--- Decision Tree Model Evaluation ---
Accuracy: 0.8596
Precision: 0.7842
Recall: 0.7885
F1-Score: 0.7863
ROC AUC Score: 0.8449
```

Random Forest

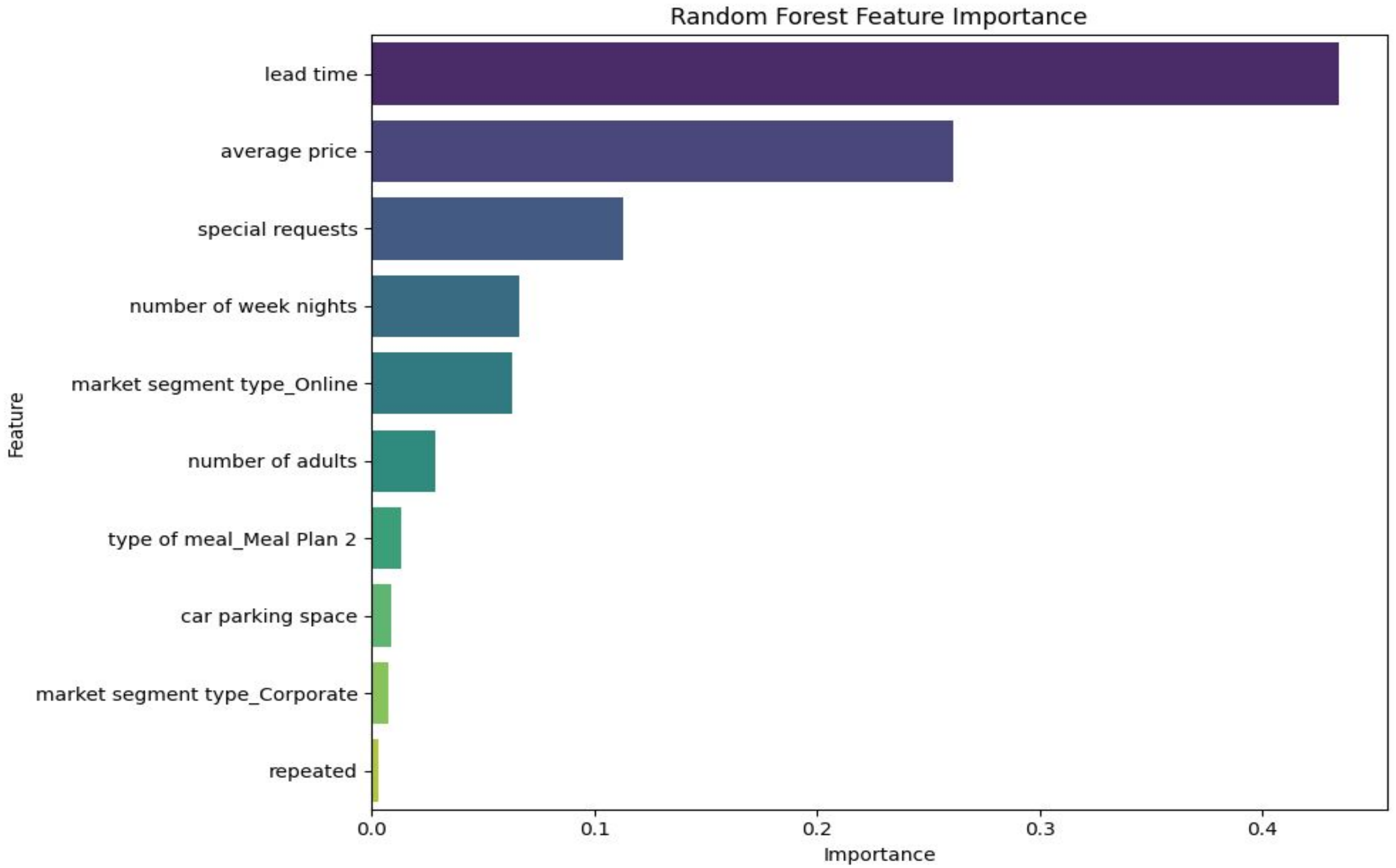
```
--- Random Forest Model Evaluation ---
Accuracy: 0.8859
Precision: 0.8507
Recall: 0.7906
F1-Score: 0.8195
ROC AUC Score: 0.9427
```

K-Nearest Neighbors(KNN)

```
--- K-Nearest Neighbors Model Evaluation ---
Accuracy: 0.8663
Precision: 0.8194
Recall: 0.7595
F1-Score: 0.7883
ROC AUC Score: 0.9120
```

5. Methodology: Feature Importance

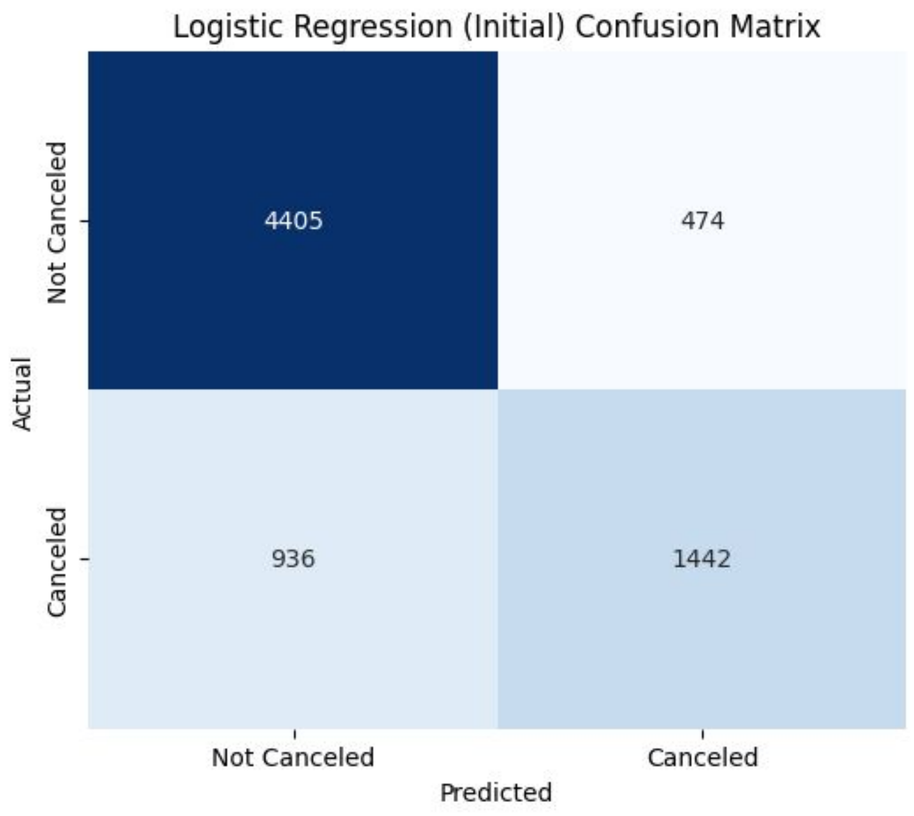
Random Forest



5. Methodology: Train Models Tuned and Confusion Matrix

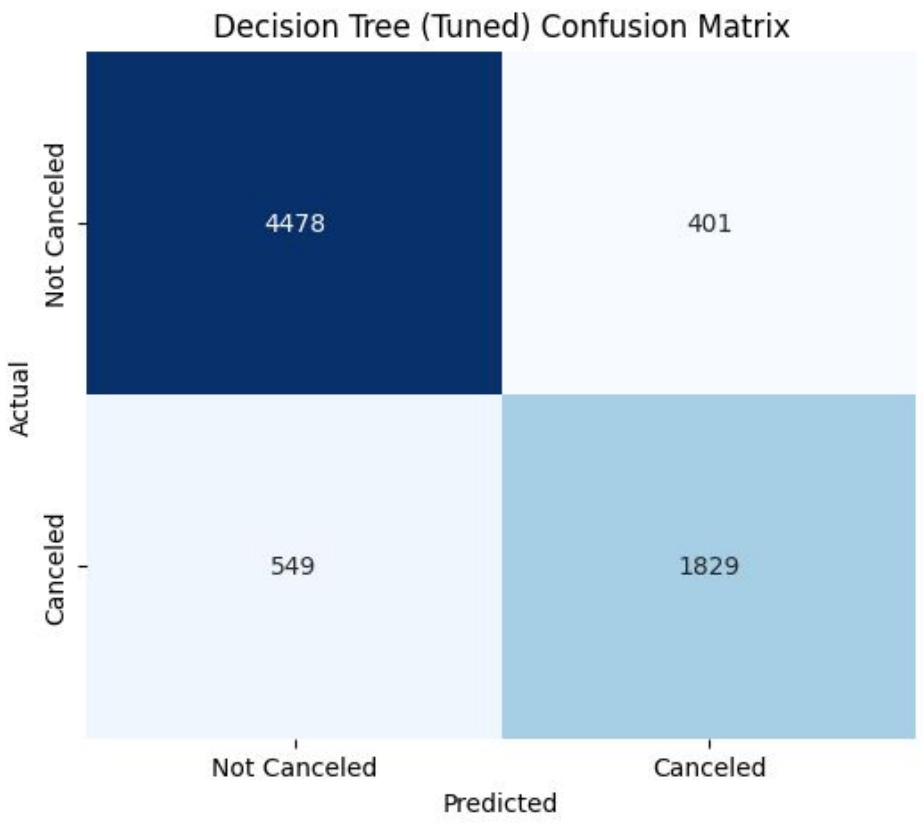
Logistic Regression (Initial Model)

- Accuracy: **0.8057 (80.57%)**
- Precision: **0.7526**
- Recall: **0.6064**
- F1-Score: **0.6716**
- ROC AUC Score: **0.8670**



Decision Tree (Tuned)

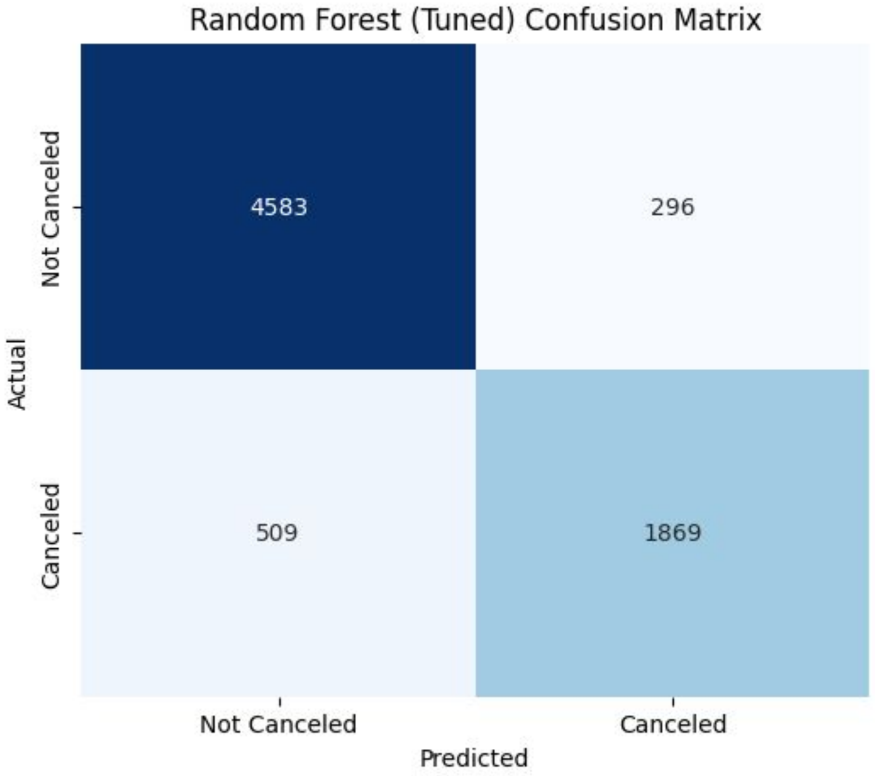
- Accuracy: **0.8691(86.91%)**
- Precision: **0.8202**
- Recall: **0.7691**
- F1-Score: **0.7938**
- **ROC AUC Score: 0.9260**



5. Methodology: Train Models Tuned and Conclusion Matrix

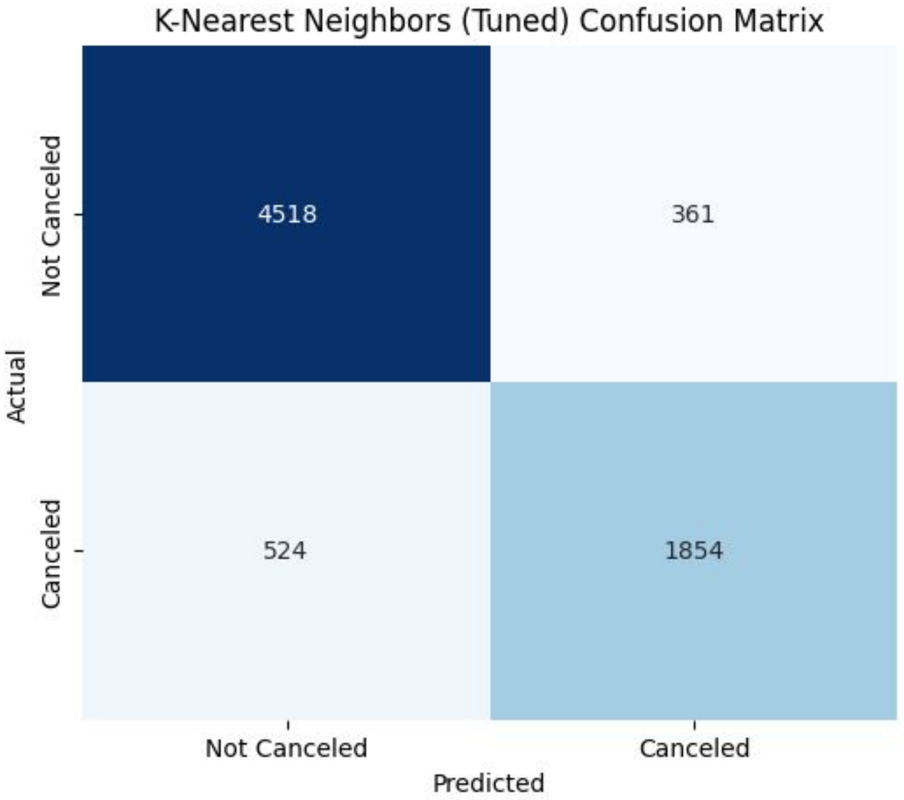
Random Forest (Tuned)

- Accuracy: **0.8891(88.91%)**
- Precision: **0.8633**
- Recall: **0.7860**
- F1-Score: **0.8228**
- **ROC AUC Score: 0.9470**



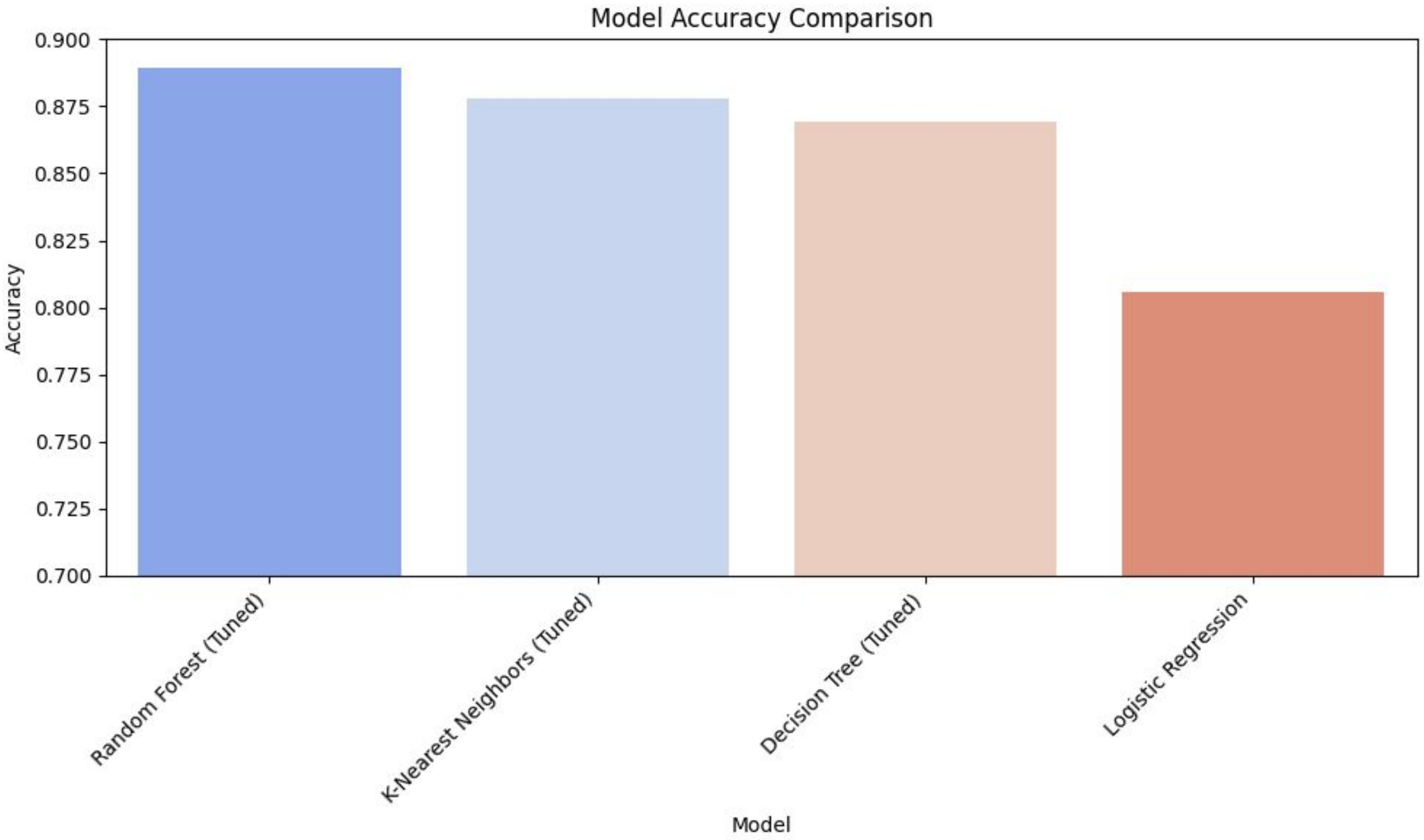
K-Nearest Neighbors (Tuned)

- Accuracy: **0.8780(87.8%)**
- Precision: **0.8370**
- Recall: **0.7796**
- F1-Score: **0.8073**
- **ROC AUC Score: 0.9263**



5. Methodology: Accuracy Comparison Bar Chart

A bar chart comparing the accuracy of all models provides a quick and intuitive understanding of their relative performance.



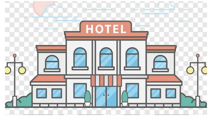
6. Findings

	Accuracy	Precision	Recall	F1-Score	ROC-AUC Score
Logistic Regression	0.8057	0.7526	0.6064	0.6716	0.8670
Decision Tree (Tuned)	0.8691	0.8202	0.7691	0.7938	0.9260
Random Forest (Tuned)	0.8891	0.8633	0.7860	0.8228	0.9470
K-Nearest Neighbors (Tuned)	0.8780	0.8370	0.7796	0.8073	0.9263

7. Conclusion

- Compared multiple machine learning models for booking cancellation prediction.
- Found the **Tuned Random Forest** to be the best-performing model.
- Achieved the highest accuracy and overall prediction performance.

Data Sources



Hotel Booking Cancellation Prediction

Source: kaggle.com

Hotel_cancellation_prediction.ipynb - Colab

Source: google_colab.com

Questions & Discussion

Thank you for reviewing the Hotel Booking Cancellation Analytics & Predictive Modeling Defense.